Han Sun, PhD

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Senior computational biologist with 10+ years at Stanford University applying AI and multiomics to uncover disease mechanisms. Builds deep learning models on genomic data and integrates human genetics with functional genomics to prioritize causal genes and therapeutic targets in diabetes, cardiomyopathy, and cancer. Author of 30+ publications and co-inventor on a treatment-methods patent, with a track record of carrying computational hypotheses through wet-lab validation.

EXPERIENCE

Consultant at a Liquid Biopsy Startup 2026 – Present

Support computational pipeline development and evaluate emerging AI models for early cancer detection.

Research Scientist / Senior Computational Biologist 2020 – Present

Department of Pediatrics, School of Medicine, Stanford University, **Prof. Anna Gloyn**

Postdoc 2015 – 2019

Department of Genetics, School of Medicine, Stanford University, **Prof. Lars Steinmetz**

EDUCATION

PhD, Bioinformatics 2009 – 2014

University of Chinese Academy of Sciences, **Prof. Yixue Li** and **Prof. Lu Xie**

Bachelor, Software Engineering 2004 – 2008

East China Normal University

SKILLS

- **Multiomics for identifying causal genes/variants and understanding disease mechanisms**: genomics (WGS, WES, arrays), transcriptomics (RNA-seq, scRNA-seq), proteomics (MS/MS), epigenomics (ATAC-seq, ChIP-seq, WGBS), immune repertoire profiling (TCR-seq/BCR-seq), HLA typing and imputation, linkage analysis, QTL mapping, and GWAS
- **Artificial intelligence for target identification and drug discovery**: machine learning (scikit-learn; linear and generalized linear models, support vector machines, random forests; cross-validation) and deep learning (PyTorch, TensorFlow; CNN, RNN, ResNet, Transformer, GPT; pre-training, fine-tuning, and transfer learning; agentic AI)
- **Programming and computing**: Python, R, C/C++, FastAPI/FastHTML, high-performance computing (HPC), and cloud computing
- **Bioinformatics and biostatistics**: BLAST, BWA, STAR, GATK, Scanpy, Seurat, Conda, Docker/Singularity, Snakemake/Nextflow; hypothesis testing, confidence intervals, power calculation, and Bayesian inference

PACKAGES AND CERTIFICATIONS

- **Python packages developed**: gap, hlarchical, omniQTL, spredle, deepSec, and dustle
- **Machine Learning Specialization** (Andrew Ng), Stanford University & DeepLearning.AI, 2022

- **Deep Learning Specialization** (Andrew Ng), DeepLearning.AI, 2022
- **Fundamentals of Deep Learning**, NVIDIA Deep Learning Institute, 2022

PROJECTS

Multiomics integration to study disease mechanisms in cancer, cardiomyopathy, and diabetes

- Identified disease-causing somatic/germline mutations, gene fusions, and splicing events (potential drug targets) from whole-genome, whole-exome, and RNA sequencing data.
- Analyzed bulk and single-cell RNA-seq/ATAC-seq, ChIP-seq/CUT&RUN, WGBS, and MS/MS data across dozens of projects to characterize disease-associated genes and pathways.
- Built unified QTL mapping pipelines across multiple omics data types (eQTL, sQTL, pQTL, and caQTL) and integrated QTL signals with GWAS data through colocalization and Mendelian randomization analyses (omniQTL).
- Led collaborations with wet-lab colleagues to validate findings using CRISPR/Cas9 editing, iPSC and mouse models, RT-PCR, Sanger sequencing, qPCR, and phenotyping assays.

Artificial intelligence to predict genetic ancestry, HLA types, and functional elements from DNA sequences

- Developed machine learning models to accurately predict genetic ancestry from genotyping data (gap).
- Implemented hierarchical, mixture-of-experts (MoE), and multi-task deep learning models for HLA imputation (hlarchical).
- Reproduced the SpliceAI model and extended it with Transformer and Hyena architectures for both splice site and usage prediction (spredle).
- Applied deep learning models to predict selenocysteine (the 21st amino acid) sites (deepSec).

PATENTS

- Methods of Treatment, Genetic Screening, and Disease Models for Heart Conditions Associated with RBM20 Deficiency. Francesca Briganti, Lars M. Steinmetz, **Han Sun**, and Wu Wei. WO 2020/092171.

PUBLICATIONS

1. Francesca Briganti, **Han Sun**, Wu Wei, Jingyan Wu, Chenchen Zhu, Martin Liss, Ioannis Karakikes et al. "iPSC Modeling of RBM20-Deficient DCM Identifies Upregulation of RBM20 as a Therapeutic Strategy." *Cell Reports* 32, no. 10 (2020): 108117, **co-first author**.
2. **Han Sun**, Huiying Yan, Kathryn Bieging-Rolett, Michelle Nguyen, William F. Mueller, Zhuanfen Cheng, Hong Zeng, Laura Attardi, Wu Wei, and Lars M. Steinmetz et al. "CDKN1A-RAB44 transcript fusion and oncogene activation in cancers." *bioRxiv* (2019), **co-first author**.
3. Seth A Sharp, **Han Sun**, K. Alaine Broadaway, Patrick E MacDonald, Anna L Gloyn, et al. "GIFT: An integrated genetic, molecular, and physiological resource for human pancreatic islets." In preparation, **co-first author**.
4. Evans-Molina, Carmella, Yasminye D. Pettway, Diane C. Saunders, Seth A. Sharp, Thomas SR Bate, **Han Sun**, Heather Durai et al. "Heterogeneous endocrine cell composition defines human islet functional phenotypes." *Nature Communications* (2026).
5. Umapathysivam, Mahesh M., Elisa Araldi, Benoit Hastoy, Adem Y. Dawed, Hasan Vatandaslar, **Han Sun**, Johanna E. Mayrhofer, Peter Lindquist et al. "Type 2 diabetes risk alleles in peptidyl-glycine alpha-amidating monooxygenase influence GLP-1 levels and response to GLP-1 receptor agonists." *Genome Medicine* (2026).
6. Rajesh, Varsha, Dora Evelyn Ibarra, Jing Yang, Haichen Zhang, Amy Barrett, Eleanor G. Kaplan, Amit Kumthekar, **Han Sun**, et al. "Functional Characterization of Glucokinase Variants to Aid Clinical Interpretation of Monogenic Diabetes." *International Journal of Molecular Sciences* 27, no. 1 (2025): 156.

7. Russ-Silsby, James, Yunkyeong Lee, Varsha Rajesh, **Han Sun**, Mahsa Amoli, Nasser Ali Mirhosseini, Tushar Godbole, Matthew B. Johnson et al. "Complete Loss of PAX4 causes Transient Neonatal Diabetes in Humans." *Molecular Metabolism* (2025): 102201.
8. de Winter, Twan JJ, Miha Sovrovic, Esmee Dekker, Natascha de Graaf, **Han Sun**, James D. Johnson, Anna L. Gloyn, Francoise Carlotti, Eelco JP de Koning, and Anna Alemany. "Cell type-specific eQTL detection from single-cell RNA-seq reveals post-transcriptional regulatory mechanisms in human islets." *bioRxiv* (2025): 2025-01.
9. Kolic, Jelena, WenQing Grace Sun, Haoning Howard Cen, Jessica D. Ewald, Jason C. Rogalski, Shugo Sasaki, **Han Sun**, et al. "Proteomic predictors of individualized nutrient-specific insulin secretion in health and disease." *Cell Metabolism* 36, no. 7 (2024): 1619-1633.
10. Ewald, Jessica D., Yao Lu, Cara E. Ellis, Jessica Worton, Jelena Kolic, **Han Sun**, Shugo Sasaki, Dahai Zhang et al. "HumanIslets.com: Improving accessibility, integration, and usability of human research islet data." *Cell Metabolism* (2024).
11. Rottner, Antje K., Yingying Ye, Elena Navarro-Guerrero, Varsha Rajesh, Alina Pollner, **Han Sun**, Romina J. Bevacqua, Jing Yang et al. "A genome-wide CRISPR screen identifies CALCOCO2 as a regulator of beta cell function influencing type 2 diabetes risk." *Nature Genetics* 55, no. 1 (2023): 54-65.
12. Lau, Hwee Hui, Nicole AJ Krentz, Fernando Abaitua, Marta Perez-Alcantara, **Han Sun**, Jun-Wei Chan, Jila Ajeian, Soumita Ghosh et al. "PAX4 loss of function increases diabetes risk by altering human pancreatic endocrine cell development." *Nature Communications* 14, no. 1 (2023): 6119.
13. Mattis, Katia K., Nicole AJ Krentz, Christoph Metzendorf, Fernando Abaitua, Aliya F. Spigelman, **Han Sun**, Jennifer M. Ikle et al. "Loss of RREB1 in pancreatic beta cells reduces cellular insulin content and affects endocrine cell gene expression." *Diabetologia* 66, no. 4 (2023): 674-694.
14. Cortez, Briana N., Hui Pan, Samuel Hinthorn, **Han Sun**, Nicola Neretti, Anna L. Gloyn, and Cristina Aguayo-Mazzucato. "Heterogeneity of increased biological age in type 2 diabetes correlates with differential tissue DNA methylation, biological variables, and pharmacological treatments." *GeroScience* (2023): 1-21.
15. Torres, Jason M., **Han Sun**, Vibe Nylander, Damien J. Downes, Martijn van de Bunt, Mark I. McCarthy, Jim R. Hughes, and Anna L. Gloyn. "Inferring causal genes at type 2 diabetes GWAS loci through chromosome interactions in islet cells." *Wellcome Open Research* 8 (2023).
16. Alghamdi, Tamadher A., Nicole AJ Krentz, Nancy Smith, Aliya F. Spigelman, Varsha Rajesh, Alokkumar Jha, **Han Sun**, Mourad Ferdaoussi et al. "Zmiz1 is required for mature β-cell function and mass expansion upon high fat feeding." *Molecular Metabolism* 66 (2022): 101621.
17. Chenchen Zhu, Jingyan Wu, **Han Sun**, Francesca Briganti, Benjamin Meder, Wu Wei, and Lars M. Steinmetz. "Single-molecule, full-length transcript isoform sequencing reveals disease-associated RNA isoforms in cardiomyocytes." *Nature Communications* 12, no. 1 (2021): 1-9.
18. Benedikt Rauscher, William F Mueller, Sandra Clauder-Münster, Petra Jakob, M Saiful Islam, **Han Sun**, Sonja Ghidelli-Disse, Markus Boesche, Marcus Bantscheff, Hannah Pflaumer, Paul Collier, Bettina Haase, Songjie Chen, Rene Hoffman, Guangwen Wang, Vladimir Benes, Gerard Drewes, Michael Snyder, Lars M Steinmetz. "Patient-derived gene and protein expression signatures of NGLY1 deficiency." *Journal of Biochemistry* (2021).
19. Semih Calamak, Menekse Ermis, **Han Sun**, Saiful Islam, Michael Sikora, Michelle Nguyen, Vasif Hasirci, Lars M. Steinmetz, and Utkan Demirci. "A Circulating Bioreactor Reprograms Cancer Cells Toward a More Mesenchymal Niche." *Advanced Biosystems* 4, no. 2 (2020): 1900139, **co-first author**.
20. Jay W Schneider, Saji Oommen, Muhammad Y Qureshi, Sean C Goetsch, Rhianna S Sundsbak, Wei Guo, Mingming Sun, **Han Sun**, Dennis A Webster, Alex W Coutts, Francesca Briganti, Wu Wei, Lars M. Steinmetz, Daniel F Carlson, and Timothy J. Nelson et al. "Dysregulated ribonucleoprotein granules promote cardiomyopathy in RBM20 gene-edited pigs." *Nature Medicine* (2020): 1-13.
21. William F. Mueller, Petra Jakob, **Han Sun**, Sandra Clauder-Münster, Sonja Ghidelli-Disse, Diana Ordonez, Markus Boesche, Marcus Bantscheff, Paul Collier, Bettina Haase, Vladimir Benes, Malte Paulsen, Peter Sehr, Joe Lewis, Gerard Drewes, Lars M. Steinmetz. "Loss of N-glycanase 1 Alters Transcriptional and Translational Regulation in K562 Cell Lines." *G3: Genes, Genomes, Genetics* (2020).

22. Arne H. Smits, Frederik Ziebell, Gerard Joberty, Nico Zinn, William F. Mueller, Sandra Clauder-Munster, Paola Grandi, Petra Jakob, Anne-Marie Michon, **Hanice Sun**, Karen Tessmer, Tilmann Burckstummer, Marcus Bantscheff, Lars M. Steinmetz, Gerard Drewes, and Wolfgang Huber. "Biological plasticity rescues target activity in CRISPR knockouts." *Nature Methods* (2019): 1-7.
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24. **Han Sun**, Chen Chen, Baofeng Lian, Menghuan Zhang, Xiaojing Wang, Bing Zhang, Yixue Li, Pengyuan Yang, and Lu Xie. 2015. "Identification of HPV Integration and Gene Mutation in HeLa Cell Line by Integrated Analysis of RNA-Seq and MS/MS Data." *Journal of Proteome Research* 14 (4): 1678–86, **co-first author**.
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28. Menghuan Zhang, Hong Li, Ying He, **Han Sun**, Li Xia, Lishun Wang, Bo Sun, et al. 2015. "Construction and Deciphering of Human Phosphorylation-Mediated Signaling Transduction Networks." *Journal of Proteome Research* 14 (7): 2745–57.
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35. Xiao Chang, Yun Li, Jie Ping, Xiao-Bin Xing, **Han Sun**, Peng Jia, Chuan Wang, Yuan-Yuan Li, and Yi-Xue Li. 2011. "EcoBrowser: A Web-Based Tool for Visualizing Transcriptome Data of Escherichia Coli." *BMC Research Notes* 4 (October): 405.
36. Xiao-Bin Xing, Qing-Run Li, **Han Sun**, Xing Fu, Fei Zhan, Xiu Huang, Jing Li, et al. 2011. "The Discovery of Novel Protein-Coding Features in Mouse Genome Based on Mass Spectrometry Data." *Genomics* 98 (5): 343–51.